

AE460 Senior Design Hybrid Rocket Technical Report 2025-2026

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This report documents the design, fabrication, and testing of “Smokey Shuttle”, a hybrid-powered rocket developed to compete in Category B of the 2026 FAR-OUT competition, targeting 45,000 feet. The vehicle uses a monotube hybrid motor that uses nitrous oxide as the oxidizer and aluminized plastisol as the solid fuel. The engine utilizes a self-pressurized blowdown system initiated through a UC valve ignition mechanism. The airframe follows a sub-minimum diameter configuration with the motor comprising the majority of the body. Two static test fires were conducted, with one resulting in a fill line failure and a full oxidizer dump, and the other producing a successful full-duration burn with recorded thrust data. The rocket is scheduled for flight in June 2026 in the Mojave Desert, California.

I. Nomenclature

<i>ARRD</i>	=	Auto Retention Release Device
<i>TWR</i>	=	thrust to weight ratio
<i>NOS</i>	=	nitrous oxide
<i>NOP</i>	=	normal operating pressure
<i>MEOP</i>	=	maximum expected operating pressure
<i>LOX</i>	=	liquid oxygen
<i>PTC</i>	=	push-to-connect
<i>GSE</i>	=	ground support equipment
<i>SRAD</i>	=	student researched and developed
<i>SSTA</i>	=	Student Space Technology Association
<i>COTS</i>	=	commercial off the shelf
<i>RTV</i>	=	room temperature vulcanizing

II. Introduction

This project serves as the capstone senior design project for thirteen aerospace engineering seniors in the Mechanical, Aerospace, and Biomedical Engineering department at the University of Tennessee, Knoxville. The team sought to design, build, and fly a hybrid rocket under the advisement of Dr. J.E. Lyne and the mentorship of Isaac Smith and Alex Van Bruggen for competition in the 2026 FAR-OUT competition. Component fabrication was performed by the team using CNC machining and composite lay-up techniques, with system testing conducted at the engine test facility located at the UTIA ETREC Plant Sciences Unit. Flight operations are scheduled for June 2026 at the Friends of Amateur Rocketry site near Mojave, CA. The project advances hybrid rocket propulsion research at UTK by contributing additional performance data for the aluminized plastisol propellant formulation developed and utilized by several preceding senior design teams. This work was carried out with support from the Student Space Technology Association, which provided fluid, control, and data acquisition ground support equipment as well as infrastructure improvements to the engine test facility. The team was organized into five subteams: propulsion, aerostructures, recovery, electronics, and simulation and aerodynamics, each responsible for their respective vehicle subsystems.

1. Team Structure:

- Owen Porteous	Team Lead & Electronics Sub-team
- Benjamin Reed	Deputy Team Lead & Electronics Sub-team
- David Nguyen	Electronics Sub-team Lead
- Ryan Finn	Propulsion Sub-team Lead
- Aidan Lankford	Propulsion Sub-team
- Avah Cherry	Propulsion Sub-team
- Cannon Emory	Propulsion Sub-team
- Jess McKinney	Propulsion Sub-team
- James Smith	Recovery Sub-team Lead
- Nick Moore	Aerostructures Sub-team Lead
- Reid Ruckman	Aerostructures Sub-team
- Aidan Crisci	Aerostructures Sub-team
- Mason Blackwood	Aerodynamics and Simulation Sub-team Lead
- Dr. J.E. Lyne	Advisor
- Issac Smith	Mentor
- Alex Van Bruggen	Mentor

III. System Architecture

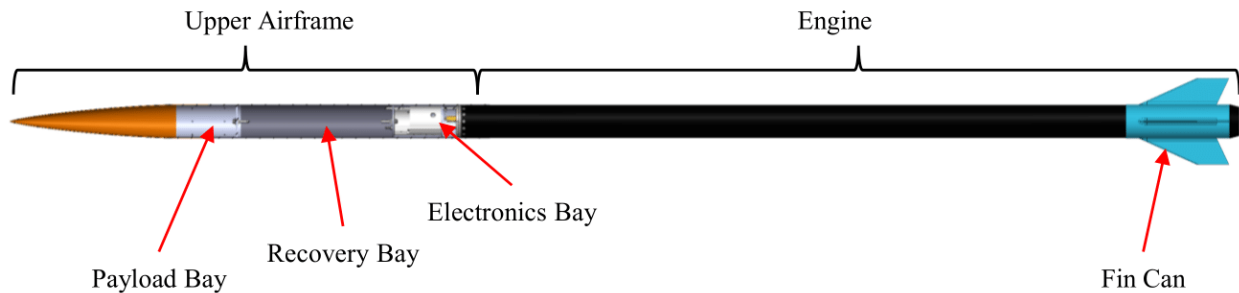


Fig. 1 Fully-Integrated Launch Vehicle

A. Propulsion Subsystems

1. Engine Architecture

The engine is a monotube hybrid rocket motor utilizing liquid nitrous oxide and solid aluminized plastisol as propellant. Oxidizer is fed into the combustion chamber via a self-pressurizing blowdown system, with a zigurat injector used to induce swirl in the nitrous flow and increase contact with the interior face of the solid fuel grains. The combustion chamber is encased within an ablative phenolic liner, used to provide an additional layer of thermal protection to the motor body tube. Fuel grains are scored and epoxied into the phenolic liner with 1/8" thick phenolic rings separating each grain, with heatsink graphite pre- and post-combustion chambers mounted at the fore and aft ends of the fuel grain assembly. These provide additional flame-holding stability forward of the combustion chamber and increase propellant mixing aft of the fuel grain assembly, respectively. A graphite nozzle is used in conjunction with an aluminum nozzle retainer, which houses the nozzle and serves as an extension of the diverging section.

Nitrous oxide was selected as the oxidizer due to its high vapor pressure, which enables self-pressurization and eliminates the need for an external pressurant system. At ambient temperatures below 97°F, nitrous oxide exists in a saturated state within a confined volume, consisting of both liquid and vapor phases. The liquid phase serves as the primary oxidizer due to its high density, while the vapor phase provides the necessary pressurization. Nitrous oxide is widely used in amateur-scale hybrid rocket engines and has been the oxidizer of choice in all previous rockets.

developed by hybrid senior design teams and SSTA rockets. However, a notable disadvantage is cost, approximately \$45 per gallon compared to \$2 per gallon for liquid oxygen (LOX). Additionally, nitrous oxide blowdown systems experience a continuous decrease in tank pressure during operation, resulting in a corresponding decline in thrust and a mixture ratio that is rarely optimal throughout the burn.



Fig. 2 Regression Rates Comparison [1]

Aluminized plastisol is a rubbery, inert plastic fuel developed by SSTA and used extensively in previous hybrid motor designs. It comprises liquid plastisol, a PVC derivative, and 20% by weight aluminum powder. While motor performance using alternative fuel types such as ABS and paraffin wax, solid fuels extensively used in amateur hybrid rocketry, was simulated in Rocket Propulsion Analysis (RPA), aluminized plastisol theoretically produced better engine performance than any other solid fuel type.

Bait Plastics 362 Saltwater liquid plastisol and Pyro-Aluminum 2-micron Indian Blackhead aluminum powder, a formulation successfully used by previous hybrid design teams, was combined and heated to 350°F in two-minute increments, then cast in a pre-heated pressurized mold to produce cylindrical solid fuel grains.

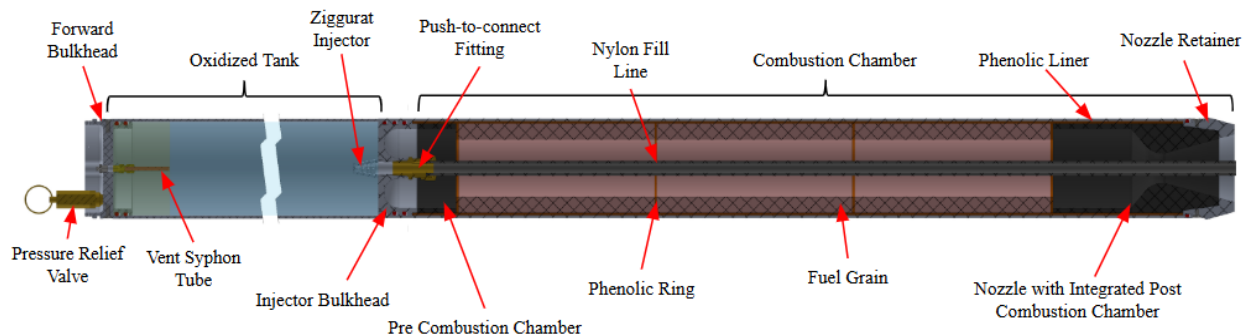


Fig. 3 Engine Cross-Section

The engine is a monotube design, in which the oxidizer tank and combustion chamber are separated by a floating injector baffle. Though damage was sustained to the aluminum body tube during the previous year's burn-through event, design modifications were made such that a shorter section of the engine casing could be re-used to minimize manufacturing costs. The monotube design is advantageous in that it avoids the use of a joint at the interface between the oxidizer tank and combustion chamber, improving rigidity and eliminating a potential point of failure. A monotube is also advantageous in that an Urbanski-Colburn (UC) valve, which consists of a push-to-connect fitting and burn-away fill line, can be utilized during ignition, eliminating the need for an actuated valve. Monotube designs

have been successfully implanted in several previous SSTA and senior design projects, including Andromeda, Redshift, and Big Hybrid.

2. Oxidizer Tank Design

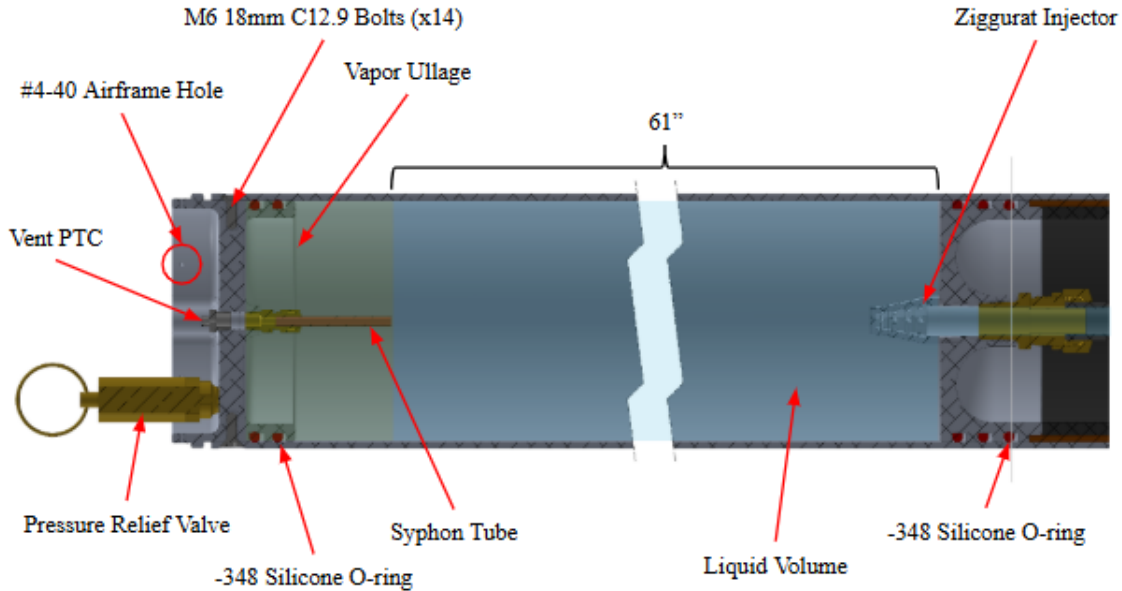


Fig. 4 Oxidizer Tank

The engine casing is a 106.4" long, 5.00" outer diameter, 0.125" wall thickness 6061-T6 aluminum radially bolted pressure vessel with a forward bulkhead and nozzle retainer, each secured with fourteen ISO class 12.9 M6 bolts and sealed with silicone O-rings. The tube, originally sourced from Online Metals, was re-used from the previous year's design. As such, the engine casing meets the minimum factor of safety (FOS) of 2 for burst against maximum expected operating temperature (MEOP), in accordance with FAR-OUT regulations. The normal operating pressure (NOP) was chosen as 750 psi, as this aligns with typical pressure values of nitrous oxide at 70°F.

Table 1 Tube Material Properties [4] [5]

Material	Aluminum
Alloy	6061
Temper	T6
Ultimate strength (psi)	45,000
Yield strength (psi)	40,000
Shear strength (psi)	30,000
Bearing yield strength (psi)	56,000

As shown in Fig. 4, a siphon tube is integrated into the forward bulkhead and is used to establish the tank fill level. The tube vents nitrous vapor from the tank through a .0292" diameter orifice, which restricts the flow rate to a negligible level under normal vapor conditions. When liquid nitrous reaches the siphon tube, the mass flow rate through the vent increases significantly due to the higher density of the liquid, producing a visible white vapor cloud that serves as an indication of fill completion. After the fill valve is closed, the tank liquid level rapidly equalizes to the height defined by the siphon tube. An active vent is essential during filling; without continuous vapor venting, the tank pressure will equilibrate with the pressure in the fill line, preventing further liquid inflow.

In accordance with FAR-OUT rules, the system must be manually depressurized. To achieve this requirement without needing to test the casing to twice the NOP, this must be accomplished remotely. The fill box that SSTA uses has a ball valve with a string attached to it that can be opened from a distance. This abort system is necessary because the launch could be put on hold at any moment during the fill sequence. It is, for this reason, disadvantageous to use

the aforementioned vent. If a plane flies overhead and the launch is postponed even momentarily, the team must abort since the rocket cannot maintain pressure for an extended period.

The tank design incorporates a single circumferential row of M6 bolts at both the forward and aft ends. This approach is consistent with the 2023 Hybrid and the 2024 Hybrid. The current design uses fourteen Class 12.9 M6 bolts, evenly spaced around the tank circumference, with a radial edge distance of 0.55 inches from the casing edge to the bolt hole center. Structural integrity was evaluated analytically by considering the primary stress modes in a bolted pressure vessel, including hoop stress, casing tensile stress, bolt tear-out, and bolt shear. Bolt shear is often a preferred failure mode, as it minimizes damage to the pressure vessel casing, which is typically the most costly component. However, bolt shear became a problem while hydro testing the body tube, discussed later, so this failure mode was given extra factor of safety. This behavior is governed by the strength, size, and quantity of the fasteners. Casing tensile strength is also influenced by bolt size and count, as larger or more numerous fasteners reduce the effective material cross-section between holes. Bolt tear-out resistance depends on both bolt diameter and edge distance, while bolt bearing failure is governed by hole geometry and represents localized deformation of the bolt interface.

Table 2 Tank Failure Mode FOS

Failure Mode	FOS
Hoop Failure	2.25
Bolt Shear	4.19
Bolt Tear-out	2.33
Casing Tensile Failure	3.08

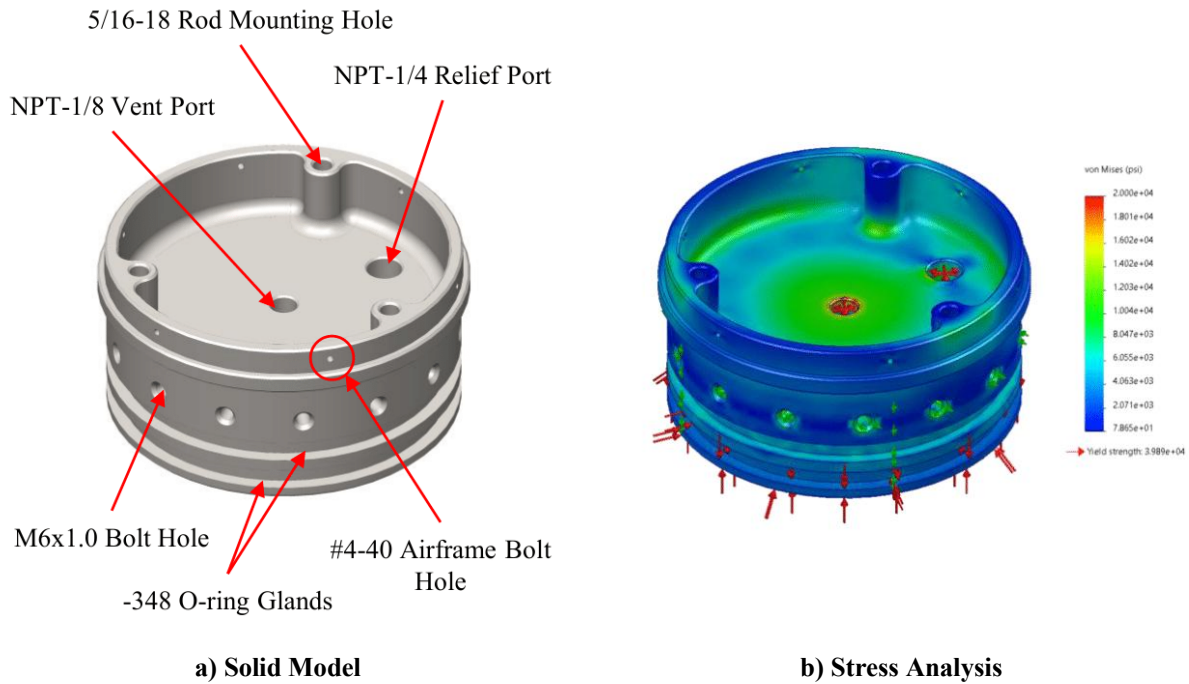
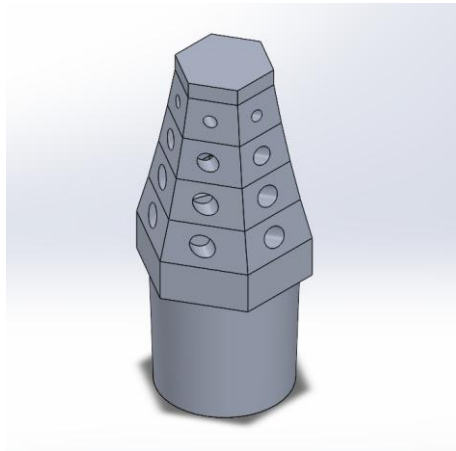


Fig. 5 Forward Bulkhead

The forward bulkhead from the 2024-25 rocket was reused. When it was designed, FEA was performed using the SolidWorks stress simulation program. The model was constrained using a roller support on the interface between the bulkhead and the body tube. Pressure forces were applied to the inside faces of the parts that contact the oxidizer fluid, including the first o-ring gland and NPT ports. The simulation was run using a pressure of 1000 psi and a fine mesh. Several design iterations were made, primarily to increase the thickness of the inside flat until the stress in the

part was less than half of the yield strength of 40,000 psi. This design took inspiration from similar bulkheads used in previous projects. Two o-ring glands were included to minimize leaking potential. Previous rockets experienced leaking issues with 200 series o-rings so this design resized the glands to fit 300 series o-rings. An external lip was added with threaded airframe attachment holes to mount an e-bay sleeve. Three mounting holes were also included for rods to transfer deployment forces to the bulkhead and engine. Fillets were added to reduce stress concentrations in the corner, and they were sized to existing tooling (3/8" ball end mill which can machine a 3/16" radius fillet).

3. Injector Design

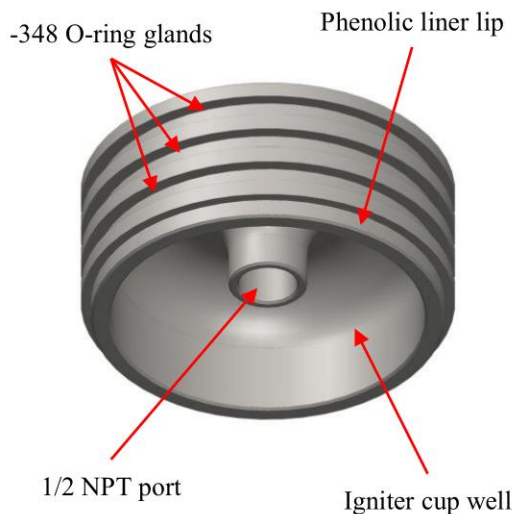


a) Solid Model

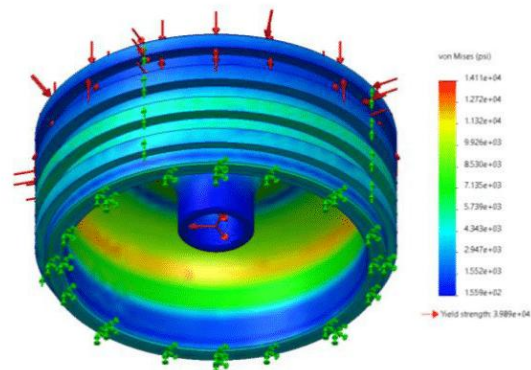


b) Physical Model

Fig. 6 Ziggurat Injector



a) Solid Model



b) Stress Analysis

Fig. 7 Injector Baffle

The ziggurat injector and injector baffle were also used from last year's rocket. The injector, shown in Fig. 6, was designed to swirl the oxidizer into the combustion section. By swirling the oxidizer into the fuel grains, the efficiency of the rocket and the consistency of the fuel grain regression is greatly improved. This design was initially proposed by a hybrid team in 2023-24. That team welded three brass fittings together to create a swirling injector. Based on their success, last year's team decided to upgrade the design to a similar, single aluminum component. Fig. 6a displays the final CAD design and Fig. 6b displays the machined part. The design is about 2.13" in length, the

ziggurat structure varies in diameter from 1.01" to 0.57", and the threaded section is an NPT size 1/2". The stock used was ~4" long when it was placed in a CNC to begin the top of the ziggurat structure. The holes along the ziggurat are all coplanar which allowed for faster machining. The holes along the injector were varied in order to maintain an equal or smaller size than the exit hole to function properly. The diameters of these holes increase from 0.07" to 0.11" to 0.13" along the top three layers before maintaining the 0.13" diameter. The holes were machined using the A axis on a CNC machine. Once the top section was complete, the bottom section was started. Since the thread pitch and diameter vary in an NPT fitting, an Excel spreadsheet was used to calculate the necessary dimensions.

4. Combustion Chamber Design

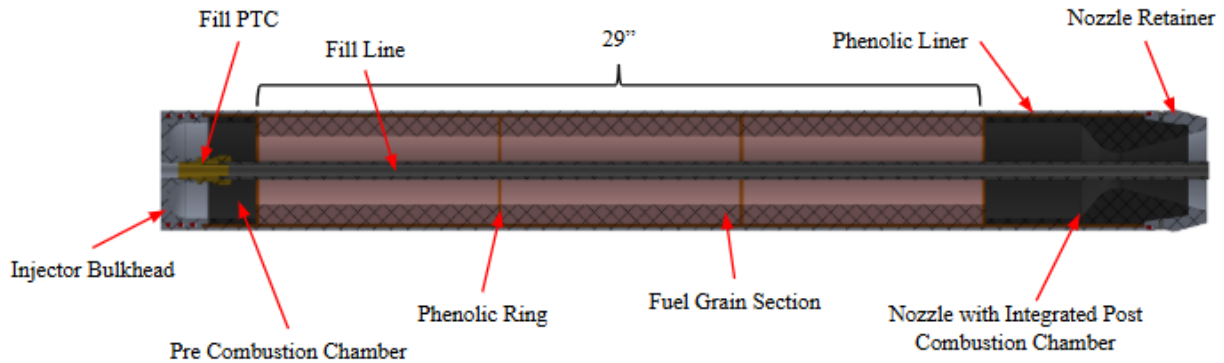


Fig. 8 Combustion Chamber Cross-Section

Since the rocket is a monotube design, the combustion chamber is contained within the same body tube as the oxidizer tank. This section is constructed within an 1/8" thick paper phenolic liner from Franklin Fibre – Lamitex to protect the outermost aluminum casing from thermal loading. Since it is an ablative material that is designed to withstand high temperatures, it also acts to resist burn-through after the fuel grain has been depleted. Within this liner, three sections of fuel grain are secured with a 2-part epoxy (Aeropoxy PR2032 and PH3665). The cylindrical grains are scored in a cross-hatch pattern on the outside with a knife to improve the grain adherence and avoid tearing. In 2024, the fuel grain on the senior design "Big Hybrid" was ejected from the nozzle after shearing. The scoring depth is approximately one quarter of the grain thickness, and lines are spaced about 1" apart from each other. The fuel portion consists of three cylindrical sections of aluminized plastisol that total 29" in length. These sections are separated by 1/8" thick Garolite CE phenolic spacers cut from sheets purchased from McMaster-Carr.

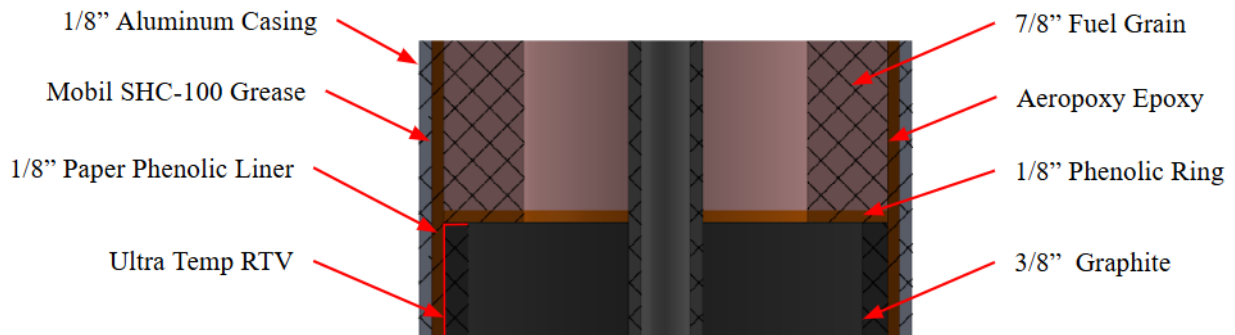


Fig. 9 Combustion Chamber Wall Cross-Section

A 2.00" long, 4.49" outer diameter heatsink graphite pre-combustion chamber is seated forward of the fuel grain assembly, allowing for more complete vaporization of the liquid oxidizer prior to contact with the solid fuel, thereby improving combustion efficiency. In contrast with previous years' designs, which used a 0.25" wall thickness, the thickness of the pre-combustion chamber was increased to 0.375" to improve durability. The nozzle geometry was

retained from the 2025 hybrid engine, incorporating a 1.50" throat, 15° half angle, and area ratio of 5.22. While the design of the combustion chamber drew heavily from previous teams' successes, including Redshift and Big Hybrid, several key design changes were made to mitigate the risk of burn-through, which caused failure of the 2025 hybrid motor. This included redesign of the post-combustion chamber, which lengthens the combustion chamber by 4" to allow for additional recirculation zones for uncombusted propellant. Ablation of the RTV interface between the post-combustion chamber and nozzle in the 2025 combustion chamber design is believed to be the primary cause of the burn-through, leaving a direct path for combustion gases to burn through the phenolic liner and damage the motor casing. To mitigate this, the post-combustion chamber and nozzle were machined as a single assembly, eliminating this seal. To mitigate risk of post combustion chamber fracture, a problem that occurred in the 2025 design, the post combustion chamber thickness was also increased from 0.25" to 0.375". To further minimize burn-through risk, an o-ring gland was incorporated into the aluminum nozzle retainer, used to seat the nozzle within the combustion chamber assembly and lengthen the diverging section, improving sealing against combustion gases. Additional changes were made to the nozzle retainer design to prevent stripping of the bolt holes, which caused premature wear in the 2025 design, and minimize risk of tearout failure. The retainer walls were thickened to allow for use of longer bolts and improved thread engagement, and the class of bolts used was upgraded from ISO 10.9 to ISO 12.9 to improve tensile and yield strength.

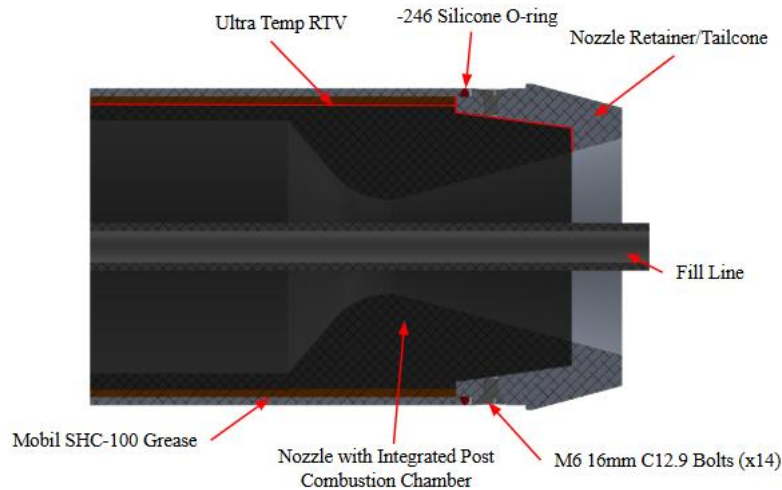
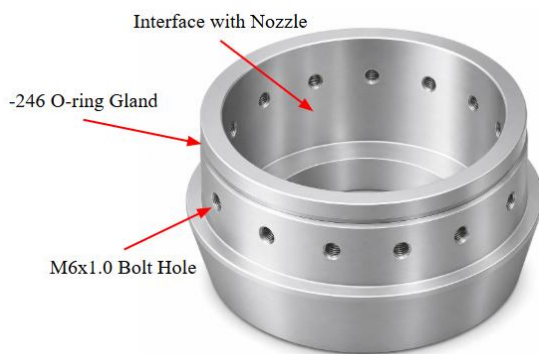


Fig. 10 Aft Section Design



a) Solid Model



b) Physical Model

Fig. 11 Nozzle Retainer

5. Engine Ignition



Fig. 12 Igniter Cup on Injector Bulkhead

With the oxidizer tank and the solid fuel grains explained, the only remaining piece needed in the combustion reaction is heat. To ignite the rocket and start its propulsion, two things need to happen: introduction of heat and disconnection of the nitrous fill line. To address both of these requirements at once, the team decided to use thermate, an incendiary pyrotechnic composition, specifically a variation of thermite, designed to generate intense heat in small, focused areas. The exact composition of thermate by weight is as follows: Iron Oxide 23%, Ammonium Perchlorate 36%, Magnalium 9%, Barium Nitrate 13%, Sulfur 2%, and Parlon 17%. The chemistry of a thermitic reaction will not be detailed in this paper. The thermate mix contains its own oxidizer and fuel and requires a high ignition temperature to start the exothermic reaction, which will be provided by e-matches. To make thermate, first weigh out each ingredient using a fine scale. Then, mechanically mix the ingredients until it becomes a homogeneous mix. To prevent any accidental reaction, perform all mixing using wooden tools. Parlon has no contribution in the chemical reaction of thermate, but it is in this mixture as an acetone-soluble binder. Mixing the thermate mixture and acetone in small batches into the igniter cup (will be discussed later), the end consistency should be similar to that of cake batter, or good pancake batter (note that the writer was most likely hungry). Thermate should take around 1-2 days to cure fully, which is not necessary for ignition. The e-matches can be placed inside before the mixture is cured, or afterwards. Ensure that the full bulb of the e-match is inserted into the thermate mixture. Secure the e-matches onto the igniter cup lid (will be discussed later) using hot glue (it's easy to work with, you can use something else). Three e-matches were used for the current design of the igniter cup, placed in a triangular pattern. They are wired in parallel onto two wires, one positive and one negative. These connections should be secure enough to not accidentally be ripped out during the rocket assembly and transportation process. The wires should be long enough to provide a safe distance from the power source to the rocket. When in doubt, be safe. The igniter cup was designed to fit into the donut-shaped cavity in the injector baffle. It sits directly below the push-to-connect and is held in place by electrical tape. A lid was made to contain the thermate, although it is not absolutely necessary, the lid prevents any broken pieces of thermate from falling out into the combustion chamber. The lid also helps secure the e-matches in place. The e-matches and extension wires are wrapped around the fill line and secured in place using electrical tape. This further prevents accidental dislocation of the wires during assembly and transport.

6. Testing and GSE

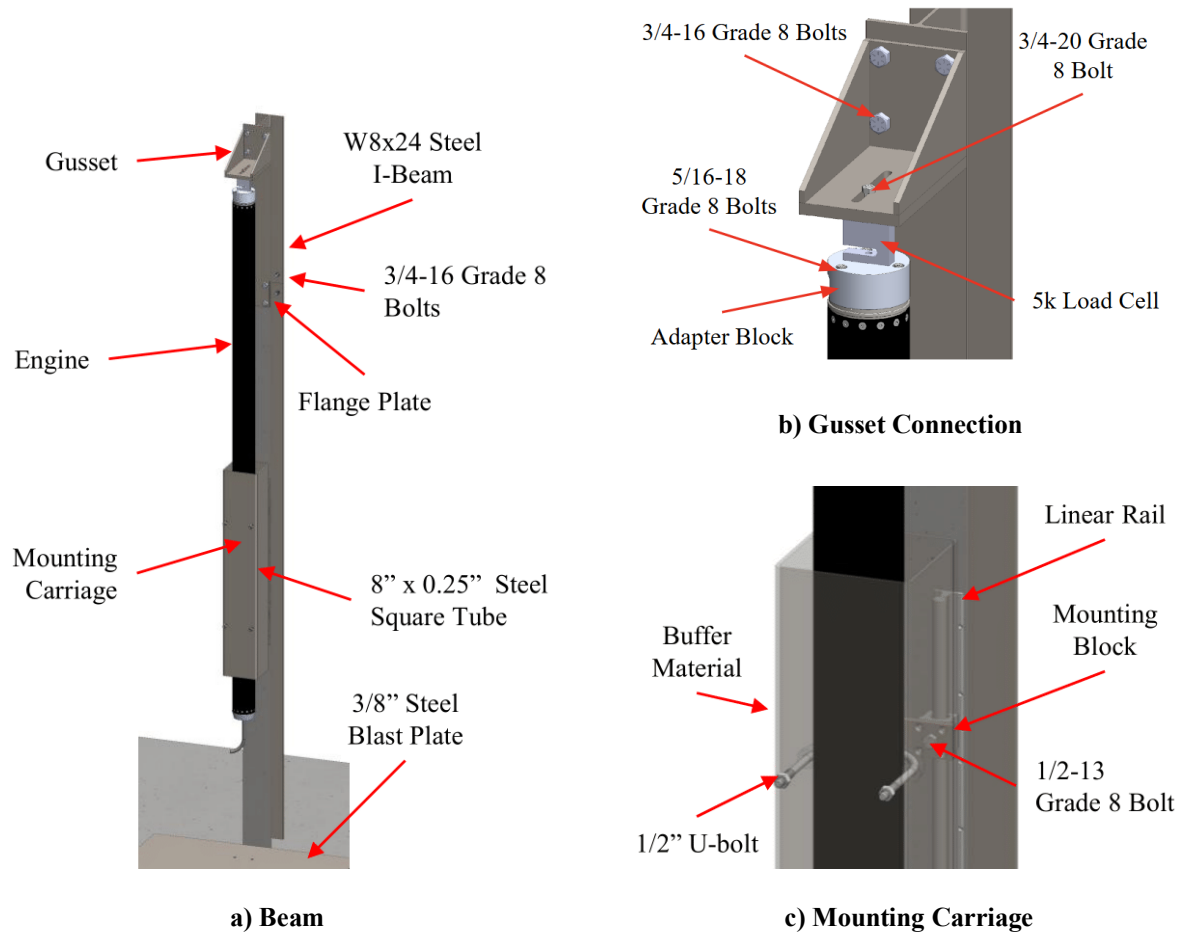


Fig. 13 Engine Test Configuration

Engine testing was conducted at the UTIA ETREC Plant Sciences facility test stand. Prior years' testing used a 3000 lb load cell that uses a 1/2-20 bolt. This year, the 3k load cells were not functional so a 5000 lb load cell was used. This required boring out the slot in the gusset at the top of the I-beam to fit a larger diameter bolt. Furthermore, the pass-through hole in the adapter block was also bored out to fit the same increase in bolt size according to the load cell. The new diameter of the bolt used is 3/4 in. The 5000 lb load cell uses 3/4" fine thread bolts, which were purchased at a local hardware store. The new hardware is only available in higher grades, starting from 8.

The engine is suspended from the thrust gusset mounted at the top of the beam using a 3/4 in diameter fine thread bolt, ideally less than 2 1/4 in long, as shown in Fig. 12(b). The custom-machined adapter block was designed to interface with the engine using the existing 5/16-inch holes. The load cell is centrally mounted within this block, and an integrated opening provides access for vent tubing and a pressure tap to the forward bulkhead.

Lateral restraint of the engine is provided by a carriage assembly that translates along linear rails attached to the beam. The carriage is fabricated from an 8-inch square steel tube with 0.25-inch wall thickness. The engine is secured within the carriage using U-bolts lined with vinyl tubing. This vinyl interface allows for radial expansion of the tank under pressurization while minimizing stress concentrations induced by the U-bolts.

B. Aerostructures Subsystems

Due to the 2024-2025 senior design hybrid rocket team's launch failure, the upper and lower airframes were able to be reused by our team. All reused parts were inspected for safety and usability prior to reuse. The lower airframe is

a 108.5" long, 5" OD aluminum tube that houses the fuel grain, oxidizer tank, and nozzle. At the opposite end of the lower airframe from the upper airframe is attached the fin can.

The fin can is composed of a 15" long, 5" ID filament wound fiberglass tube, to which are connected 4 carbon fiber fins. The fiberglass tube, AKA the "can," was purchased from Madcow rocketry and is connected to the lower airframe using the same screws that hold the nozzle retainer in place. The dimensions of the fins are as follows; root chord: 10", tip chord: 5.5", semi-span: 4", sweep length: 7", thickness: 0.2". The base of the trailing edge of the fins was offset 2" from the aft end of the can. These dimensions were chosen to suit the aerodynamics of the entire rocket using OpenRocket such that a stability caliber of 1.5-1.8 would be achieved. As it turned out, the length of the lower airframe was reduced shortly after the fins were made due to a failed pressure test, so the payload had to be adjusted to account for the change in length.

The first step in creating the fins was to cut 17 identical squares of carbon fiber alternating 0° and 45°, ensuring that 4 fins of the dimensions given above would fit on each square. The alternating directions of cuts ensure that the carbon fiber filaments are at 45° relative to each other, increasing the strength of the final plate. The 17 plates are then used to create a plate using the VARTM (Vacuum assisted resin transfer molding) process using Aeropoxy PR2032 epoxy. This process was performed with the assistance of UTK's *SSTA* Big Liquid team, whose help was greatly appreciated. After the VARTM plate was made, a table saw and band saw were both used to cut out fins of the specified dimensions. The leading edges of the resulting fins were beveled on a belt sander, and then aligned with the rotational axis of the fin can using the 3D printed jigs in Figure 14.

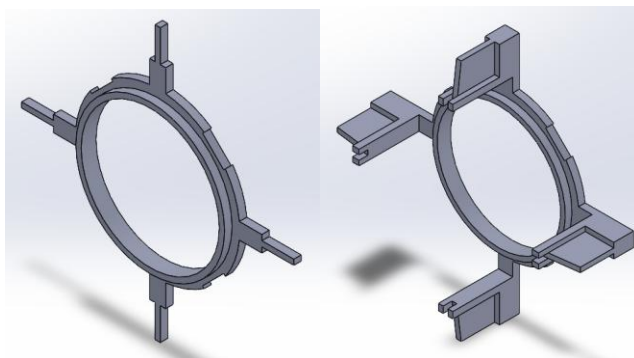


Fig. 14 Fin Alignment Jigs for Fin Can

The fins were tacked to the fin can using Aeropoxy PR2032 epoxy and the jigs above to keep them straight. After the epoxy cured, fillets of Aeropoxy ES6209 epoxy were added to both sides of the bases of each fin to ensure stability at high speeds. A heat gun was used to remove a majority of the bubbles in the fillets, as the ES6209 epoxy tends to aerate significantly during mixing. Finally, a tip-to-tip layup process was used to lay on 5 more layers of carbon fiber in increasing size and alternating 0° and 45° angles between each fin. The epoxy used for the layup process was Totalboat High Performance epoxy. Finally, the complete fin can was sanded and evened out, and finally holes were drilled and countersunk at the aft end of the fin can for attachment to the lower airframe using the same M6 bolts that hold on the nozzle retainer. The close-to-final product can be seen in Figure 15.



Fig. 15 Fin Can Assembly

The upper airframe is a 5" ID filament-wound fiberglass tube purchased from Madcow Rocketry that houses the electronics bay, recovery bay, and oxidizer tank vent tube. The upper bulkhead acts as the interface between the upper and lower airframes and is attached to the upper airframe with M6 bolts. A hole with rounded corners was cut into the side of the bottom of the upper airframe, and a hole 0.75" smaller than the upper airframe hole, also with rounded corners, was cut into a 5" OD fiberglass tube. A plexiglass window was heated and pressed around the fiberglass tube such that the ID of the plexiglass was the same as the OD of the tube. The tube was then slid into the upper airframe and glued into place with Aeropoxy ES6209 2-part epoxy such that the 2 cutouts in the tube and upper airframe were concentric. The plexiglass window was then cut to size to fit into the hole in the upper airframe and rest on the outer face of the now fixed fiberglass tube. 5/32" Holes were drilled through the plexiglass and the fiberglass tube. Twelve 0.172" #4-40 brass threaded inserts for plastic were sanded down to the thickness of the inner fiberglass tube and inserted into the holes in the tube. Finally, 12 #4-40 screws were used to secure the plexiglass window in place. The final result can be seen in Figure 16.



Fig. 16 Plexiglass Window in Upper Airframe

The total upper airframe assembly can be seen in Figure 17. The electronics bay sits on three 5/16-18 threaded rods inserted into holes in the upper bulkhead. The threaded rods extend up through an upper shield plate made from offcuts from the VARTM fin plate made earlier. Attached through holes in the plate is a U-bolt for the parachute to attach to. Above this shield plate is the recovery bay which houses the parachute and its deployment system. The shield plate must be airtight in order for the deployment charge to be effective. Finally, the payload bay is inside the coupler tube and nosecone.

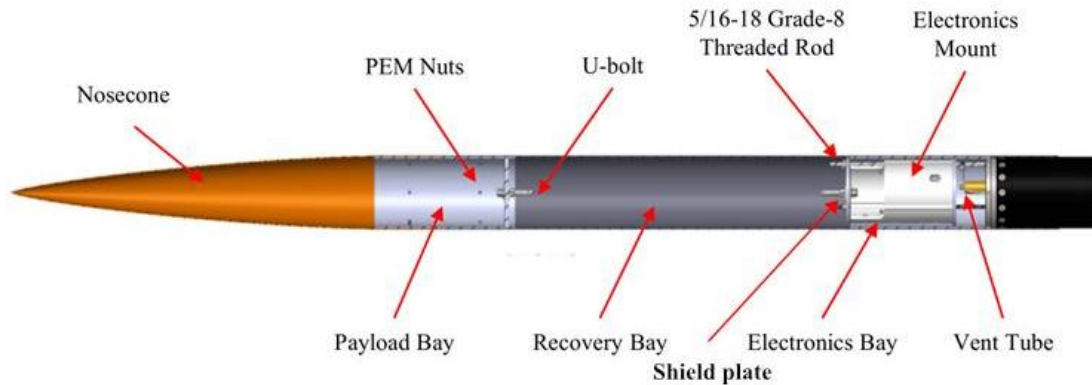


Fig. 17 Upper Airframe Assembly with Nosecone

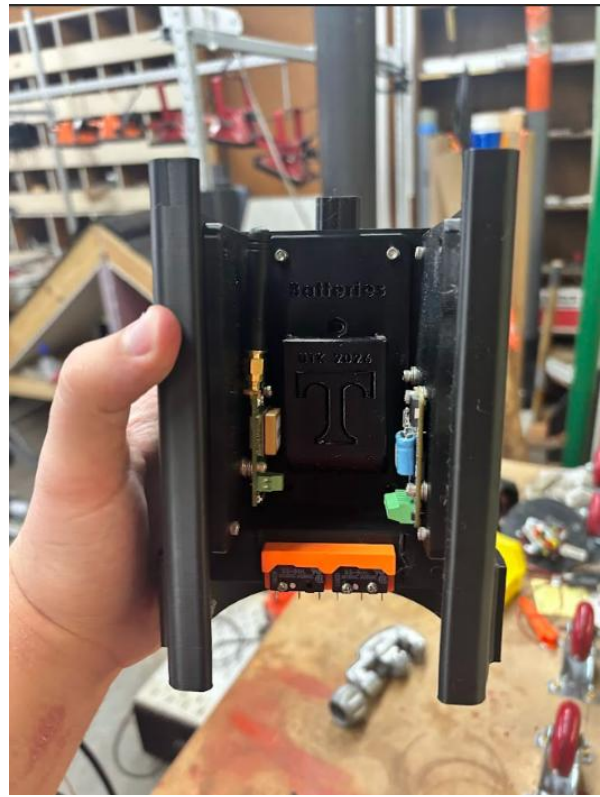
The nosecone was reused from the previous hybrid rocket team's design as it had also sustained no damage from the previous team's misfire. The nosecone houses the payload and is connected to the upper airframe via a 10" coupler tube with a 5" overlap on both the nosecone and upper airframe. The nosecone connects to the coupler tube via #4-40 screws, while the coupler connects to the upper airframe via nylon screws inserted into PEM nuts in the coupler tube. These nylon screws, AKA shear pins, shear off once the parachute charge detonates, launching the nose cone off and dragging the parachute and drogue chute out after it.

C. Electronics Subsystems

The electronics subteam is responsible for the avionics bay, data acquisition, and recovery deployment systems for the flight. The electronics chosen for the subsystem are two Blue Raven altimeters, 1 Featherweight GPS Tracker, and a double pull pin switch from Inverted Pursuits Lab. The batteries chosen are three 3.7 V LiPos, a 600 mAh for the GPS and two 500 mAh for the altimeters. The electronics bay consists of five 3D printed parts designed in SolidWorks and fabricated by the team: the housing, GoPro mount, and three sleds.



a)



b)

Fig. 18 Electronics Bay Dry and with Avionics

The electronics bay is a completely redesigned assembly compared to last years, replacing the cylindrical sled approach with a flat-plate modular design. The primary housing structure and GoPro mount were printed out of ABS plastic for heat resistance and structural rigidity. The part went through multiple redesigns and 3 physical prototypes before becoming the final product seen in Fig. 18a. Initial designs had three triangular faces, with the electronics mounted on the outside of the walls. Throughout redesigning, the structure became three rectangular walls to allow space for the electronics and GoPro, became thinner to fit inside the upper airframe coupler, obtained built in nut holes for front-face assembly, and shed extra material that added weight and hindered wiring capabilities. The final housing contains an open front face for 2 reasons: to allow the GoPro to sit inside of the electronics bay and face out of a side window in the rocket, and for easy accessibility to the electronics from this window. The electronic components are fully accessible and modifiable through this window without the need of a full disassembly of the upper airframe, a considerable improvement in serviceability compared to previous designs.

The electronics are mounted on three flat plate sleds, printed out of TPU for vibrational damping, which are mounted on the inside of the electronics bay. Two of the sleds are used to hold the GPS and altimeters, which are labeled with the outlines of each for easy assembly. These sleds were designed in a way that allows for the two altimeters to be mounted on one, while the GPS can share the same mounting screw holes and be installed on the other. These sleds contain screw and nut holes that allow the electronics to be mounted on, using washers as spacers to protect the circuit board having direct contact with the sled. The other sled is a separate design, and features three slots to hold the LiPo batteries, along with a set screw mount over top to hold them in place, as seen in Fig. 19. The sleds are mounted onto dedicated screw holes in the main housing structure using #4-40 screws. This allows for easy removal of the sleds with an allen key, giving ease of access to the electronics without needing to work within the cramped interior.

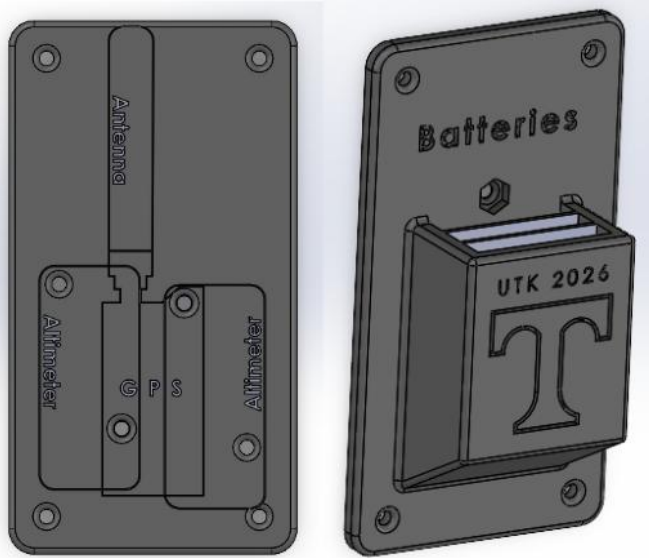


Fig. 19 Sleds for Electronics and Batteries

A custom GoPro mount, printed out of ABS, was designed and is placed directly on top of the electronics bay housing. This mount was designed to hang the camera upside down within the inside of the housing. This allows for the GoPro and electronics to be accessible from the same window, while minimizing the amount of space taken up by the electronic systems in the rocket. The mount is triangular shaped with a hole in the middle to minimize weight, and it has a standard GoPro mounting bracket underneath which is shown below in Fig. 20. This bracket allows for official GoPro screws, nuts, and housings to be used. The compact design of the electronics bay prevents the screw from coming out while assembled, but the housing can be rotated and the camera can be taken out easily of the housing, making this a non-issue.

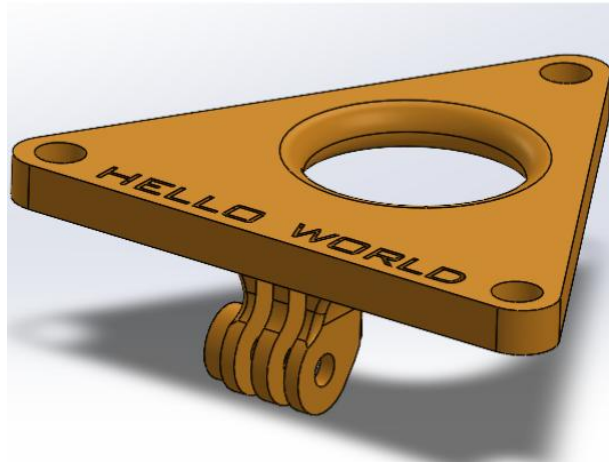


Fig. 20 GoPro Mount

The electronics, particularly the altimeters used for recovery, are armed using a double pull-pin switch kit purchased from Inverted Pursuits Lab. The switch uses two Omron SS-5GL micro-switches mounted in a bracket with a single 4" stainless steel pin. Pulling the pin simultaneously closes both circuits, arming the system at the pad. These circuits are wired individually, each with their own power sources. Each altimeter and GPS are powered with a Lithium polymer (LiPo) battery that outputs 3.7 V, with a charge capacity of ~500 mAh. LiPo batteries were chosen for their

lightweight, compact design and also for their voltage outputs. The pin can be inserted on either side of the device, allowing flexibility in rocket orientation while on the launch pad. An optional "Remove Before Flight" flag was included to ensure the system is armed before flight. This design replaces the key switches used in previous years, simplifying the arming procedure to a single pull action and reducing the number of airframe penetrations required.

Equally as important as going up, going down properly was a big consideration for our team. To deploy the recovery system in a timely manner, a doubly-redundant trigger system was set up, using two separate Blue Raven altimeters. The double redundancy mitigates the risk of an improper deployment of the recovery system, or worse, a complete failure of the deployment in the first place. The figure below shows how the altimeters are wired to the drogue parachute and the main parachute. The recovery system will be explained in more detail in the recovery system, so this is a brief explanation to preface how electronics will work in tandem with the recovery system.

The recovery system contains two main parts: the drogue and the main parachute. The drogue is deployed at the rocket's apogee, where it is meant to slow down and stabilize the rocket before the deployment of the main. Once the rocket reaches the deployment altitude of the main at 3000 ft AGL, the main is then deployed. For relevance to the electronics, there are two sets of e-matches that light the black-powder charges to deploy the parachutes. Each set contains two e-matches, wired to its own altimeter for redundancy. The e-matches to the drogue and the main are wired in parallel from the battery. The other sides of the e-matches are connected to the main and apogee ports on the altimeter, respectively. To activate the e-match, the altimeter closes the circuit from the port, and this fires the e-match. Each altimeter is wired identically. This means that as long as one altimeter works, the recovery system will be deployed. The power source to the altimeter is wired in parallel with the e-matches from the battery. The negative terminal of the battery is connected to the ground port on the altimeter. This double redundancy has a hidden cost. Since each altimeter can fire independently, this practically doubles the risk of a premature activation of the recovery system. This has not been mitigated in the current design.

The GPS' purpose is to track the trajectory of the rocket and its resting position after the launch. This allows the team to find the rocket and recollect it. The GPS is powered by a 600 mAh LiPo battery, which is simply wired to the positive and negative ports of the GPS. The GPS has a built-in switch on its circuit board, which will be turned on before the launch.

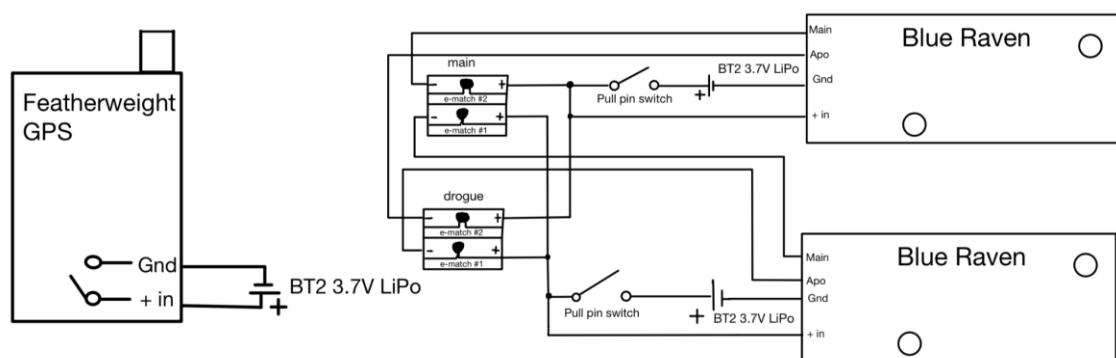


Fig. 21 Wiring diagram of the Avionic Systems

D. Recovery Subsystems

The recovery subsystem consisted of parachutes, a recovery harness, and deployment devices interfacing with the electronics subsystem to initiate deployment. The design of the recovery subsystem was a single-bay, dual-deploy system. This design contained the parachutes, recovery harness and deployment devices entirely within a single bay to reduce complexity, weight, and size. This single-bay system required a deployment device to restrict the deployment

of the main parachute. This differs from a dual-bay system where the main parachute is contained within a separate compartment and the deployment is actuated by separation; these systems require a device to actuate the deployment. There are a variety of methods to achieve deployment; common devices include cable cutters, the Jolly Logic, ARRD and the Tender Descender.

Previous failures of the ARRD system led to the selection of the Tinder Rocketry Tender Descender 3 (TD3) as the main parachute restriction device. The TD3 uses a dual-electric match actuation, allowing for each match to be wired to an independent altimeter; this allows for single-component redundancy as the ignition of one electric match (e-match) will cause the ignition of the other. The TD3 is also simpler to assemble and disassemble than the ARRD, as instead of using epoxy as a sealant the TD3 utilizes a pliable putty to secure the e-matches and create a gas-tight seal. The TD3 is mounted directly on the electronics bay bulkhead adjacent to the recovery U-bolt. Nosecone separation and drogue deployment are initiated by the detonation of a 4.5 gram black powder well charge. The well was constructed [9] from a three-quarter inch CPVC cap, and sealed by using epoxy to affix a Gatorade cap to the open end. These low-cost parts allow for the charge well to be virtually disposable and easily replaceable. Two five-sixteenth diameter holes are drilled into the CPVC cap to allow for the insertion of two separate e-matches, which can be wired to independent altimeters for redundancy.

The upper bulkhead of the electronics bay was water-jet cut out of carbon fiber, with the intention of being able to withstand the deployment of the recovery system without harming the sensitive electronics stored below. This includes holes for the U-bolt, e-match wires, and pass-through of the threaded rods connecting the upper and lower bulkheads, all sealed using plumbers putty to create a barrier for the pressure produced by the deployment charge. Due to the location of the electronics bay, the threaded rods needed to transfer the force from parachute deployment alongside the recovery bay to the lower bulkhead, which was achieved by mounting the electronics bay on these rods, disconnected from the forces of the recovery system. These rods were sized to withstand the maximum theoretical deployment force of the main parachute as prescribed in FAR-OUT rules (5.6.1.1) [2]. The strength of this assembly is determined by the tensile strength of the rods, pull-out strength of the threaded holes and the shear strength of the fiberglass plate attaching the recovery system to the nosecone. Recovery hardware was also sized in accordance to these rules, 3,000 lbf quicklinks and swivels were used on connections between the main parachute and bulkhead plate and 1,200 lbf quicklinks were used on connections between the nosecone, drogue parachute, and TD3.

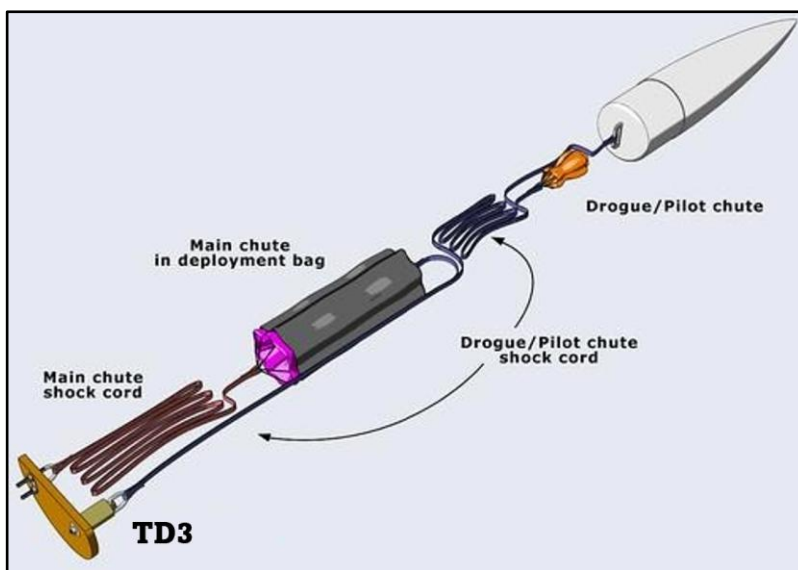


Fig. 22 Recovery System Example Layout

The parachutes for the rocket were sized to hit descent rate requirements prescribed in the FAR-OUT rules (5.6) of between 50 and 150 ft/s above 30,000' and a ground hit velocity of ~18 ft/s. Working with OpenRocket the typical sizes of Rocketman brand parachutes were input until these requirements were met. A 48 inch diameter Rocketman Pro Experimental drogue parachute and a 10 foot diameter Rocketman High Performance main parachute were selected, with coefficients of drag of 0.97 and 2.2 respectively.

E. Aerodynamics/Simulation Subsystems

The primary responsibility of the aerodynamics team was to evaluate the flight performance, stability behavior, and aerodynamic design changes required for the rocket to safely and effectively reach the target apogee. This involved developing and maintaining the full-vehicle simulation in OpenRocket, analyzing the effects of wind, launch conditions, payload mass, fin geometry, and external body modifications on the rocket's trajectory and stability margins. The goal of this work was to ensure that the final vehicle configuration satisfied the FAR-OUT competition requirements while also improving the overall flight reliability compared to the previous design. The final assembly is shown at the end of this section.

Trajectory analysis was performed using OpenRocket. After assembling the full rocket model and importing the required thrust profile, launch conditions were applied to match the expected environment at competition. The FAR-OUT rules report a suspected average ground wind speed of approximately 5 mph; however, simulations were also performed at higher speeds to evaluate sensitivity to launch conditions. Within the OpenRocket "Flight Simulations" tab, parameters such as launch rail length, wind speed, temperature, atmospheric pressure, and launch-site latitude and longitude were defined. These conditions allowed the team to predict the rail exit velocity, stability at launch rail clearance, maximum altitude, maximum velocity, angle of attack behavior, and overall trajectory performance.

OpenRocket estimates aerodynamic stability using the extended Barrowman method and performs flight propagation using a six degree of freedom (DOF) Runge-Kutta 4 numerical integration scheme. The extended Barrowman method provides estimates of the center of pressure based on the aerodynamic contributions of each component. The model accounts for translational and rotational motion making it useful for evaluating how the rocket responds to disturbances such as wind, thrust, and changes in aerodynamic loading throughout flight. This made OpenRocket a powerful tool for identifying design changes that improved stability without increasing drag or reducing apogee.

A major focus was determining how subsystem dimensions and mass distribution affected vehicle stability. The FAR-OUT competition requires the rocket maintain a minimum static stability margin of 1.5 calibers at launch rail exit for wind conditions up to 20 mph, while also achieving a rail exit velocity of at least 100 ft/s. These requirements are critical because the rocket must leave the rail with enough velocity for the fins to generate restoring aerodynamic forces. If the rocket is under-stable at rail clearance, it may not correct itself after a disturbance. If the rocket is over-stable, it becomes more susceptible to weathercocking, where the rocket rotates into the wind and loses vertical altitude performance. The two most significant design factors affecting our rockets stability were the fin can and the payload mass. The team inherited an outer vehicle configuration from the previous year. Because much of the outer structure had already been manufactured, the aerodynamic design process focused on improving the existing configuration. The original fin can was relatively thick, heavy and had limited effective planform area. Initial flight simulations showed a peak stability margin of approximately 10 calibers with the original fin can. While this technically represents a stable rocket, this level of over-stability can be undesirable because it increases the tendency of the rocket to weathercock in windy launch conditions, reducing the likelihood of reaching the target apogee.

To improve the aerodynamic behavior, the fin can was redesigned to provide a more consistent stability margin throughout flight. The updated fin design reduced excessive stability while still maintaining adequate restoring capability at launch rail clearance. With the redesigned fin can, the maximum stability margin was reduced to approximately 7.40 calibers for the 10 mph wind case. Although a typical design guideline is to keep the stability margin near 1–2 calibers, the maximum stability in this case occurred at a velocity of approximately 1,400 ft/s, where the rocket had already achieved significant aerodynamic authority. Because of this, the high peak stability margin was less concerning than the stability behavior immediately after rail departure, where the rocket is most vulnerable to wind-induced rotation. To further evaluate flight stability, I analyzed the rocket's angle of attack as a function of time.

Plotting angle of attack is important because static stability margin alone does not fully describe the vehicle's dynamic response. A rocket may have a positive static stability margin but still experience large oscillations if the aerodynamic damping is insufficient. For this reason, the simulation results were also used to evaluate damping behavior by applying a custom expression for damping ratio in OpenRocket and plotting it over time. This helped determine how quickly oscillations decayed after disturbances and whether the rocket returned to a stable flight orientation after rail departure and during high-speed flight.

Payload selection was another major part of the stability and trajectory analysis. The team selected sand as an adjustable payload material and purchased sandbags capable of holding up to 22 lb. The expected payload range was approximately 2.2–15 lb, depending on launch-day wind conditions and the final stability requirement. This adjustable payload strategy gave the team a practical method for tuning the center of gravity before launch. Because the rocket is naturally bottom-heavy due to the motor casing, phenolic liner, fuel grains, oxidizer system, bulkheads, and aft structural components, additional mass in the nosecone moves the center of gravity forward. Moving the center of gravity forward increases the distance between the center of gravity and center of pressure, which increases the static stability margin. The payload also provided a way to tune apogee performance. Increasing payload mass improves stability but reduces maximum altitude due to the added weight. Decreasing payload mass improves altitude performance but can reduce stability, especially in higher wind conditions. Therefore, the final payload mass must balance the stability requirement, rail exit velocity, wind sensitivity, and target apogee. As team lead, I used OpenRocket simulations to evaluate multiple payload cases so the team could select a launch-day payload configuration based on the expected wind conditions.

Another aerodynamic improvement involved the addition of a two-inch transition between the aluminum body tube and the fin can. This transition was added to reduce drag associated with the sudden change in body diameter between the two sections. A sharp step or exposed face between body sections can create separated flow and increase pressure drag, especially at high speeds. By smoothing this interface with a transition, the external flow remains more attached, reducing the drag penalty from the diameter mismatch. In addition to reducing drag, the transition shifted the center of pressure slightly forward and improved the predicted altitude by approximately 3,000 ft in the simulation.

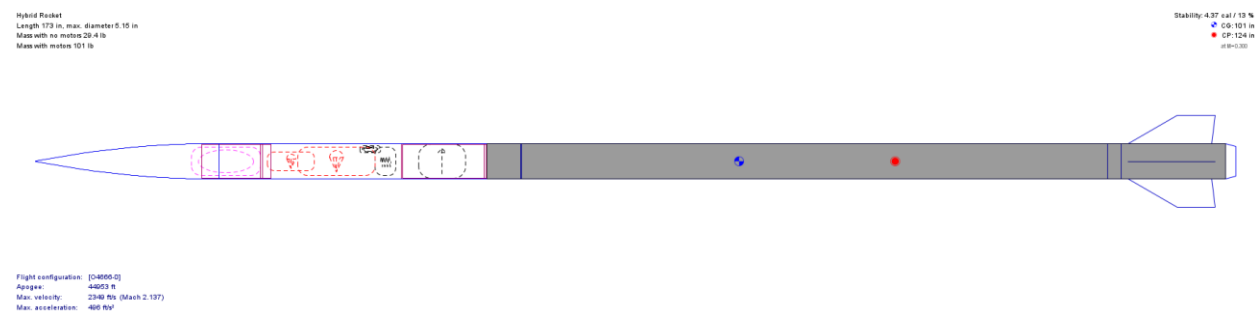


Fig. 23. OpenRocket Design

IV. Conclusions and Lessons Learned

This project was very successful, and the goal of fixing the issue with the burn through from the previous year's test was accomplished. With a successful test fire, the rocket is planned to launch at FAR OUT 2026. A third test fire is planned with the goal of getting better thrust results. This will help with accurately predicting apogee altitude and lead to a better understanding of the vehicle. Future improvements on the rocket are to create better mechanisms for assembling the propulsion system and fixing the issue where not all the fuel was burned during the test fire. Current plans to reconcile this are in the works. Future teams working on this project should make sure to maintain and improve documentation on the vehicle.

Appendix

A. System Weights, Measures, and Performance Data

1. Rocket Characteristics

Table 3 Rocket Characteristics

Stages	1
Vehicle length (ft)	15
Airframe diameter (in)	5.15
No. of fins	4
Fin semi-span (in)	4
Fin tip chord (in)	5.5
Fin root chord (in)	12
Fin thickness (in)	.3
Vehicle weight (lbm)	65.04
Propellant weight (lbm)	46.4
Empty motor weight (lbm)	65.04
Payload weight (lbm)	10
Liftoff weight (lbm)	101
Center of pressure (in)	124
Center of gravity (in)	101

2. Propulsion Information

Table 4 Propulsion Information

Propulsion type	Hybrid
Manufacturer	SRAD
Impulse class	O
Avg. thrust (N)	4666
Total impulse (N-s)	34344
Burn time (s)	8

3. Rocket and GSE Diagrams

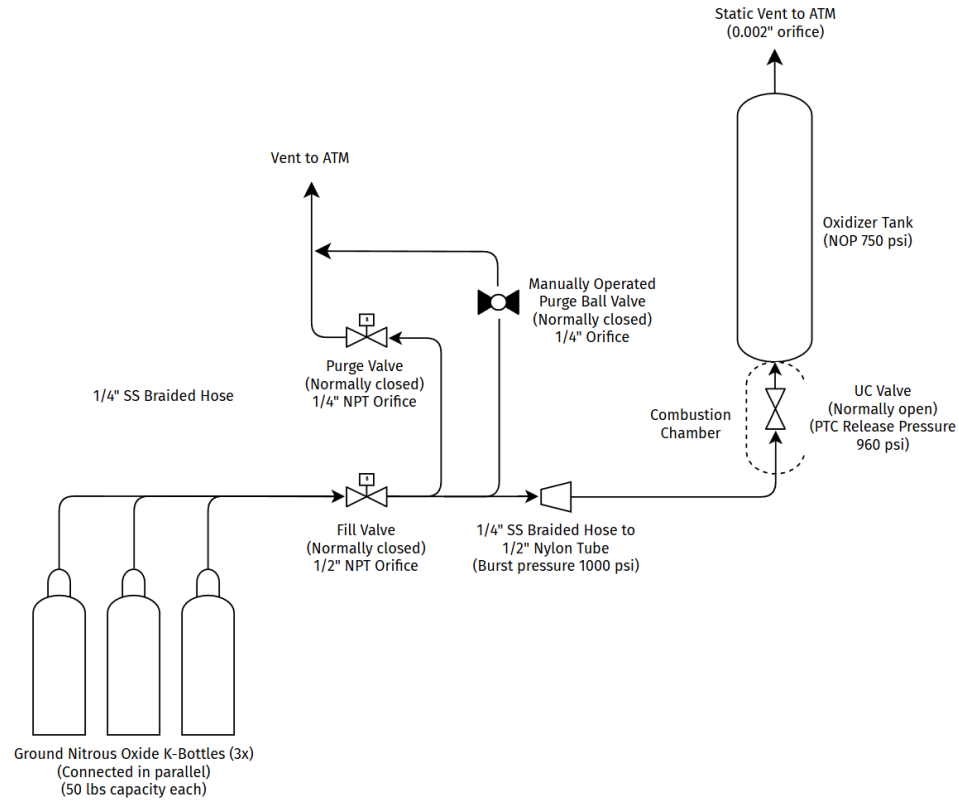


Fig. 24 Rocket/GSE P&ID

4. Predicted Flight Data

Table 5 Predicted Flight Data

Rail length (ft)	20
Liftoff TWR	16.26
Rail departure velocity (ft/s)	141
Min. static margin	1.76
Max. acceleration (G)	10
Max. velocity (ft/s)	2041
Fin flutter velocity (ft/s)	-
Target apogee AGL (ft)	45,000
Predicted apogee AGL (ft)	45,033

5. *Recovery Information*

Table 6 Recovery Information

Primary Altimeter	Featherweight Blue Raven
Redundant altimeter	Featherweight Blue Raven
Primary tracker	Featherweight GPS
Drogue deployment	CPVC Black Powder Well Charge
Main deployment	Tinder Rocketry Tender Descender 3
Drogue deploy alt. (ft)	45,000
Drogue descent rate (ft/s)	68.9
Main deployment alt. (ft)	3,000
Main descent rate (ft/s)	18.3
Shock cord rating (lbf)	6,000
Swivel rating (lbf)	3,000
Quick link rating (lbf)	2,200

B. Project Test Reports

1. Isolated Push-to-Connect and Fill Line Hydro Tests

A major concern of the rocket that stemmed from last year's attempted static fires was the reliability of the push-to-connect and fill line components. Last year's team suffered five failed attempts to complete a static fire at the hands of this system. The team tested several variations and found success using 1/2" push-to-connects from Contrail Rockets and 1/2" black nylon tubing from McMaster Carr. Though this tubing was only rated for 250 psi, the team successfully used it for a test under a load of ~750 psi.

Taking these suggestions into consideration, our team decided to purchase these parts and run isolated hydro tests to confirm the viability of the system. Four tests were conducted and the details of which can be seen in Table 7 below.

Table 7 Fill System Hydro Tests

Test Number	Test Conditions	Ideology	Results
1	Nitrogen gas with ice bath	Nitrogen is cheaper than NOS and was on hand. The ice bath simulated the temperature of flowing NOS	Pressure regulator did not work; shot tubing with 1500 psi and it burst immediately
2	Nitrogen gas with ice bath	See Test 1	Withstood 850+ psi (100 more than NOP)
3	Nitrous between k-bottles	Test the tubing performance with flowing NOS	Withstood flowing NOS at a pressure of ~500 psi
4	Nitrous between k-bottles (one with heating blanket)	Use a heating blanket to increase the pressure to a level we might see at competition	Withstood flowing NOS at a pressure of ~710 psi

2. Body Tube Hydro Tests

Our team used the same body tube from last year's rocket. However, because the aluminum body tube experienced burn through, a new hydrostatic pressure test needed to be performed after cutting the end off. New M6 countersunk bolt holes were drilled as close as possible to the end of the tube using the shop's mill. The clamp that was used limited how close this could be to the end of the tube so after the new holes were drilled, the end of the tube had to be further trimmed to reach a distance of 0.55" from hole center to the end of the casing.

The hydro test was performed by inserting the injector bulkhead to the tube and holding it in place at the bottom of the rocket against the nozzle retainer. The NPT hole in the injector baffle was plugged, and the tube was filled with water from the lake at the test stand. Though this worked, we recommended that future teams use the hose at the front facility and buckets. The forward bulkhead was then attached and nitrogen was used to pressurize the system. The tank held pressure until ~1135 psi when the forward bulkhead shot off the body tube. Some of the bolts sheared and some of the bolts stayed intact and ripped parts of the body tube out. This can be seen below in Figures 25 and 26.



Fig. 25 Bolt Failure from First Hydro Test



Fig 26. Bolt Tear Out in the Body Tube

We proposed that the mode of failure from the initial test was a stress concentration at the interface between the bolt heads and the thin aluminum at the innermost part of the countersunk holes. The design was adjusted such that the bolts were countersunk into the upper airframe and fin can rather than the body tube. This ensured that the stress from the tank pressure would be distributed to the bolts across the entire casing thickness rather than the sharp edge as designed previously.

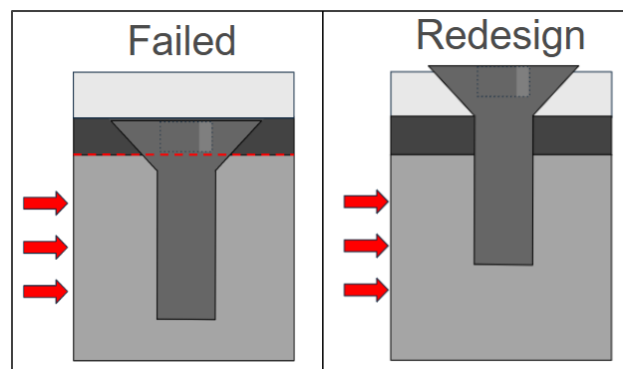


Fig 27. Bolt Tear Out in the Body Tube

Thus, the second hydrostatic pressure test was performed with M6 through holes that were not countersunk in the body tube. For the hydro test, washers were used to simulate the space between the bolt heads and the body tube that would be occupied by the upper airframe at the top of the engine, and the fin can at the bottom. This test procedure was otherwise the same as the first, and the tank held 1440+ psi for 20 minutes. This fulfills the FAR-OUT pressure testing requirements for the body tube to hold a pressure of 1.5 x MEOP for at least 15 minutes.

3. *Static Fire Testing*

Two static fire attempts were conducted with this rocket. The first attempt was made on 4/10/26 but during the oxidizer fill, the nylon tubing burst. Though the reason for the rupture is unclear, we suspect that this was the result of a stress concentration at a damaged point in the line because the push-to-connect and nylon tubing system was tested extensively in isolation without issues. These components were also the same units that last year's hybrid team used and recommended (Contrail Rockets push-to-connect and 1/2" McMaster Carr black nylon tubing with a working pressure of 250 psi). Before the hot fire test, the fill line was likely damaged during the insertion of the combustion assembly. The first assembly was inserted using a mallet. Seeing the issues this could have caused, the second assembly used a ratchet system. The ratchet provided more even pressure and eliminated the impact that could have caused the damage in the first assembly. The successful test was performed on 4/17/26.

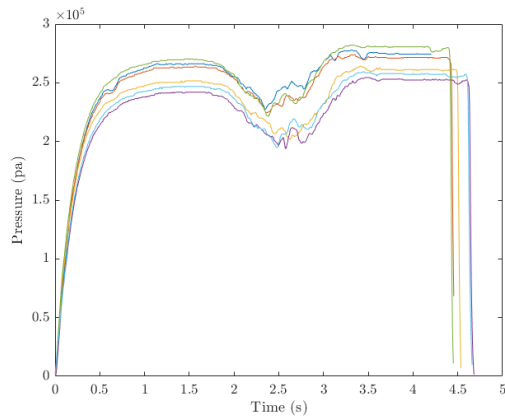
The data from this fire was bad, however, due to recording issues. The live data that was captured has several time skips and one of them happens during the start of the fire. Because of this, the peak thrust and up to several seconds of fire time was not recorded. The system also recorded and stored a more detailed report of the load cell data. However, this dataset does not record timestamps and could also be reporting incomplete data from the fire. After lots of analysis was done on this data, we concluded that even if this data were missing sections of thrust data, our thrust was significantly low. We believe this was limited by the amount of oxidizer and upon future testing, we will evaluate the performance of the engine with a greater amount of oxidizer and another method of determining max fill (in addition to the typical method of monitoring weight measurements from the load cell).

4. *Water Flow Testing*

This section was done by the 2024-2025 team, however it is very important to the design so their testing is shown here. The ziggurat injector was tested using the existing SSTA water flow test rig. The injector was located on the underside of the top plate of the rig, inside the water reservoir. A piston is located within the tank, compressed air is fed into the tank and drives the piston to pressurize the water and spray it out of the injector element. Six tests of the injector were run with pressure of the reservoir as well as the mass of water inside the tank before and after runs being measured. These tests could be improved with the inclusion of the push to connect fitting to more accurately model the restrictions present in the engine.

Table 8 Water Flow Testing Data

Run	Initial Mass (g)	Final Mass (g)	Δm	C d
1	25456	17518	7.938	0.673
2	25458	17536	7.922	0.638
3	25582	17618	7.964	0.648
4	25528	17464	8.064	0.650
5	25604	17608	7.996	0.638
6	25502	17570	7.932	0.634
Avg.	-	-	-	0.641



a) Pressure Curves



b) Spray Pattern

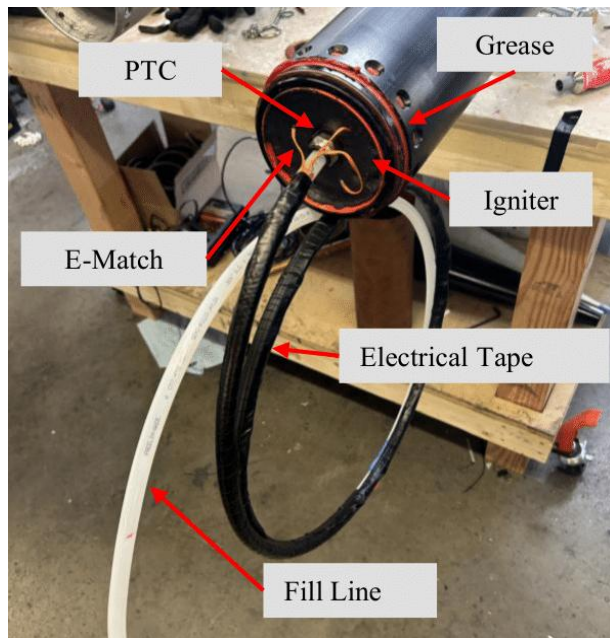
Fig. 28 Waterflow Testing

C. Fabrication/Assembly

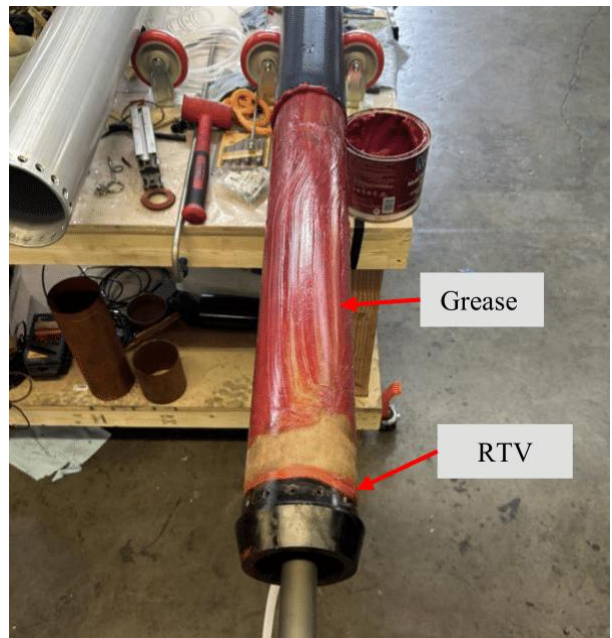
1. Engine Assembly

First the fuel grain assembly is prepared in the phenolic liner. The fuel grains are initially scored on the outer surface using a cross-hatch pattern to a depth of $3/8$ " to enhance adhesion with the epoxy. Following this preparation, the grains are then epoxied inside the phenolic liner with Aeropoxy PR2032+PH3665. For the initial test fire in 2026, phenolic spacer rings are installed between individual grain segments as well as both the top and bottom ends of the assembly. For subsequent test firings, it is recommended to test without the top and bottom phenolic rings to see if there is an increase in regression rate of the fuel grains. Clearance is intentionally maintained above and below the grain stack to accommodate the pre- and post-combustions chambers and nozzle. After the epoxy has fully cured over 24 hours, the assembly is cut to the required length and sanded to ensure proper fit with the pre-combustion chamber and nozzle.

Thorough cleaning of all engine components, in particular those in direct contact with the oxidizer, is the first step in engine assembly. The interior of the tank is cleaned with isopropyl alcohol to remove any oils or residue that may cause fouling during combustion. The forward bulkhead, injector baffle, O-rings, fittings, dip tube, and nozzle retainer are also thoroughly cleaned with isopropanol. Following cleaning, the oxidizer tank is assembled, with appropriate PPE used to prevent contamination of engine components. Krytox GPL-206, a high-temperature performance lubricant, is applied to the threads and O-ring gland of the injector bulkhead. The dip tube and O-rings are mounted, and the bulkhead is inserted into the tank. Krytox is then applied to the threads of the injector baffle, and a $1/2$ " push-to-connect fitting is mounted to the aft end. O-rings are greased using Mobil SHC-100 aviation grease and seated within the baffle O-ring glands. The injector bulkhead is now inserted into the body tube. This may require ratchet straps to push o-rings past the screw holes without damaging their sealing capability. It is necessary to complete this step prior to attaching the fill line so that the line is not damaged in assembly. Once the injector baffle o-rings are past the body tube holes, the baffle should slide smoothly by hand. Now, the fill line is secured to the push-to-connect fitting, and the ignitor assembly is mounted within the injector baffle using Permatex Optimum high-temperature RTV. E-matches are buried within the thermate, with the wiring wrapped along the fill tube using electrical tape. RTV is also applied to the face of the injector baffle that interfaces with the fuel grain assembly.



a) Injector Bulkhead System



b) Fuel Grain Assembly

Fig. 29 Engine Assembly

Now the nozzle and precombustion chamber are stalled into the phenolic liner with the fuel grain assembly. RTV is applied to all mating surfaces of the pre-combustion chamber and nozzle/post-combustion chamber prior to their insertion into the grain assembly. The nozzle and pre-combustion chamber are then installed into the liner, allowing the graphite faces to seat against the fuel grains (or in our test the phenolic rings) to form a seal with the RTV. The nozzle retainer is now installed onto the mating surface on the nozzle which is coated in RTV. Allow the retainer to seat against the phenolic liner to form a seal with the RTV here as well. Once all the graphite pieces are installed with RTV, Mobil SHC-100 grease is applied to the upper one third of the grain assembly. The fill tubing is then fed through and out of the nozzle and the grain assembly is aligned with the injector baffle and inserted into the tube. A ratchet strap may be required to assist in drawing the assembly past the nozzle retainer o-ring and fully into position. During insertion, grease should be applied continuously to facilitate smooth movement. Finally, the bolt holes are aligned, and the retainer is secured in place using bolts. Allow the RTV 24 hours to fully cure.

2. Fabrication of Combustion Chamber Parts

In previous years, machining the pre and post combustion chambers has been a very challenging aspect of building the propulsion system. Graphite is a very brittle material so an abundance of caution has to be taken when using standard shop tools. Last year the hybrid team used a CNC machine to make pilot holes for a drill press with a 4" diameter hole saw bit to hollow out graphite stock. They ran into several problems that ultimately resulted in cracked post combustion chambers and misaligned holes. This cost the team more money when they needed to buy a second round of graphite stock. To alleviate these concerns, this year's team opted to pay the nozzle machinist to make a new nozzle with the post combustion chamber attached, and to make a new pre combustion chamber. This ultimately saved the team time and money.

D. Bill of Materials

This is a non-exhaustive list of critical COTS components.

Table 9 Bill of Materials

Item	Cost	Quantity	Total
Blue Raven Altimeter	208.59	2	417.18
Featherweight GPS	177.87	1	177.87
Plasticol 5 Gal	145	2	290
Contrail PTC Component	30.24	4	120.96
25-ft Nylon Tubing	56.69	2	113.38
Graphite Rod	488.31	1	488.31
Ring Terminals	7	1	7
Respirators	40	2	80
TD-3 Tether Release	205	1	205
GoPro Hero4 Session Camera (GF)	48.7	1	48.7
FAR-OUT Flight Fee	600	1	600
Contrail Rockets PTCs	30	5	150
Pin Stop Device	25.68	1	25.68
MMCarr O-Rings	23.31	5	116.55
Wire Cutter	7	1	7
Nozzle Protective Case	43.34	2	86.68
Nozzle Graphite Machining	525	1	525
TD-3 Extra Material	20.5	1	20.5
Ignition Chemicals	57	1	57
Phenolic Plates	20.85	3	62.55
NOS Filling	1260.2	1	1260.2
Flash Drive (GF)	19.65	1	19.65
Avionics Batteries (GF)	8.73	4	34.92
Avionics Screws	14	1	14
Tailcone Transition Tube	76.56	1	76.56
Nitrogen Tank	49.71	1	49.71
Fiberglass Tube	126.09	1	126.09

EBAY Threaded Rods	27.98	3	83.94
Oval Quick Links	5.18	8	41.44
Steel Eye Nuts	8.18	4	32.72
Eye-to-eye Swivels	33.45	4	133.8
MMcarr Shipping+ Tax	40.22	1	40.22
WD40 Tub	30	1	30
Shear Pins	10	3	30
Shock Cord (per yd)	4.74	34	161.16
Rocketeer Fees (GF)	65	9	585
AirBNB (GF)	1111.67	1	1111.67
Brass Tapping Inserts (GF)	33.14	1	33.14
Sand bags	40	1	40

Acknowledgments

We would like to acknowledge the members of SSTA and Dr. Lyne who helped us greatly throughout this process. Isaac Smith was very helpful for building the basis of the electronics bay and recovery system, and was our go to person for help with any part of the rocket. Mika Deters was very helpful for the structures team, helping with creating the fin can. Alex Van Bruggen supported us in the beginning. He worked on the rocket in the year before us, so was very knowledgeable on its design and helped get us started. Ryan Mellons was helpful for machining our parts as well as general questions and test site specifics. Dr. Lyne was there throughout our process and supported us with his knowledge and expertise. Landon Meazell and Jake Ramsey were helpful with the ground system and data acquisition, being very knowledgeable and lending a hand on the technical aspects and operation of the equipment. Thank you to everyone else who supported us on this rocket as well.

We would like to also thank all of the individual donors who contributed to the GoFundme campaign that we started, through which over \$4000 was raised from 36 different donors. These funds covered a considerable amount of production and travel expenses, enabling 9 team members to attend the FAR-OUT competition. Future senior design teams are strongly encouraged to pursue crowdfunding in their projects, as it can really expand flexibility in expenses and allow for more resources to be purchased towards the rocket.

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